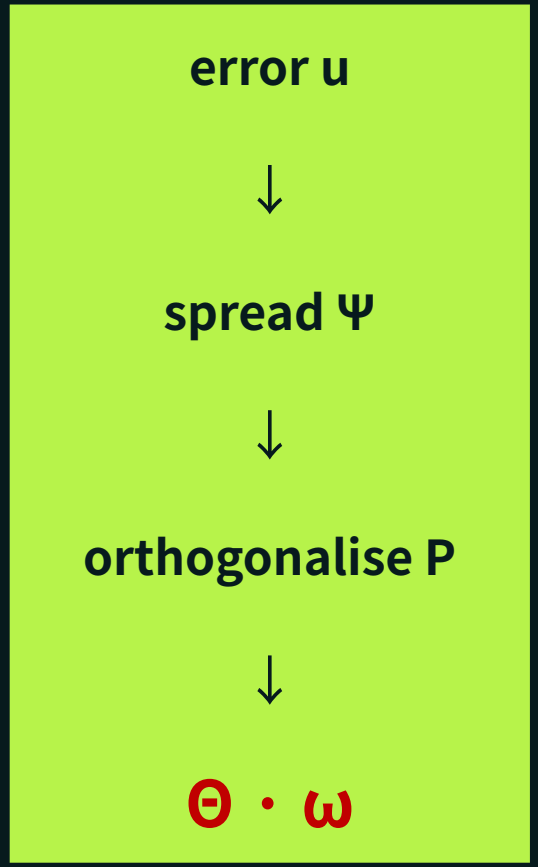
IRF and FEVD Workbook
A VAR(1) Worked by Hand

Impulse responses and variance shares, step by step



Financial Time Series · Week 6 Workbook

IRF traces how a shock spreads; FEVD shows where forecast errors come from

Two questions asked after fitting a VAR, answered with the same ingredient Θ

IRF · Impulse Response Function

How shocks spread

- Hit one shock today
- Track each variable h periods on

FEVD · Variance Decomposition

Where errors come from

- Variance of the H-step error
- Split it into shares by shock

Fit the VAR → **MA form Ψ** → **Orthogonalise P** → **IRF Θ** → **FEVD ω**

This workbook carries one example through all five steps by hand

Example: a two-variable VAR(1) in which US rates lead Korean rates

Illustrative numbers chosen for hand calculation · y_1 = US rate, y_2 = Korean rate

Model · A is the coefficient matrix

$$\begin{bmatrix} y_{1t} \\ y_{2t} \end{bmatrix} = \begin{bmatrix} 0.5 & 0 \\ 0.5 & 0.5 \end{bmatrix} \begin{bmatrix} y_{1,t-1} \\ y_{2,t-1} \end{bmatrix} + \begin{bmatrix} u_{1t} \\ u_{2t} \end{bmatrix}$$

Error covariance · same-day correlation 0.6

$$\Sigma_u = \mathrm{Var}(u_t) = \begin{bmatrix} 1 & 0.6 \\ 0.6 & 1 \end{bmatrix}$$

Reading the numbers

- Top-right 0: Korean rates do not move US rates
- Bottom-left 0.5: US rates pass to Korean rates one period later
- Diagonal 0.5: each rate keeps half of itself
- 0.6 in Σ: the two errors move together on the same day

These numbers are all we need to get every IRF and FEVD value by hand

01

Impulse Response Functions

Steps 1–6 · how one shock spreads over time

Step 1: written as a sum of past errors, the response weights are powers of A

Moving-average form: today's y is a weighted sum of today's and past errors

MA form · obtained by repeated substitution

$$y_t = u_t + A\,u_{t-1} + A^2 u_{t-2} + \cdots = \sum_{h \geq 0} \Psi_h\, u_{t-h}, \qquad \Psi_h = A^h$$

Squaring A by hand · rows times columns

$$A^2 = \begin{bmatrix} 0.5 \cdot 0.5 + 0 \cdot 0.5 & 0 \\ 0.5 \cdot 0.5 + 0.5 \cdot 0.5 & 0.5 \cdot 0.5 \end{bmatrix} = \begin{bmatrix} 0.25 & 0 \\ 0.5 & 0.25 \end{bmatrix}$$

Spot the pattern · go up to the cube

$$A^3 = \begin{bmatrix} 0.125 & 0 \\ 0.375 & 0.125 \end{bmatrix} \quad \Rightarrow \quad A^h = 0.5^h \begin{bmatrix} 1 & 0 \\ h & 1 \end{bmatrix}$$

The diagonal halves each period (0.5^h); the bottom-left grows then fades as $h \cdot 0.5^h$

One rule gives the response at every horizon without a calculator

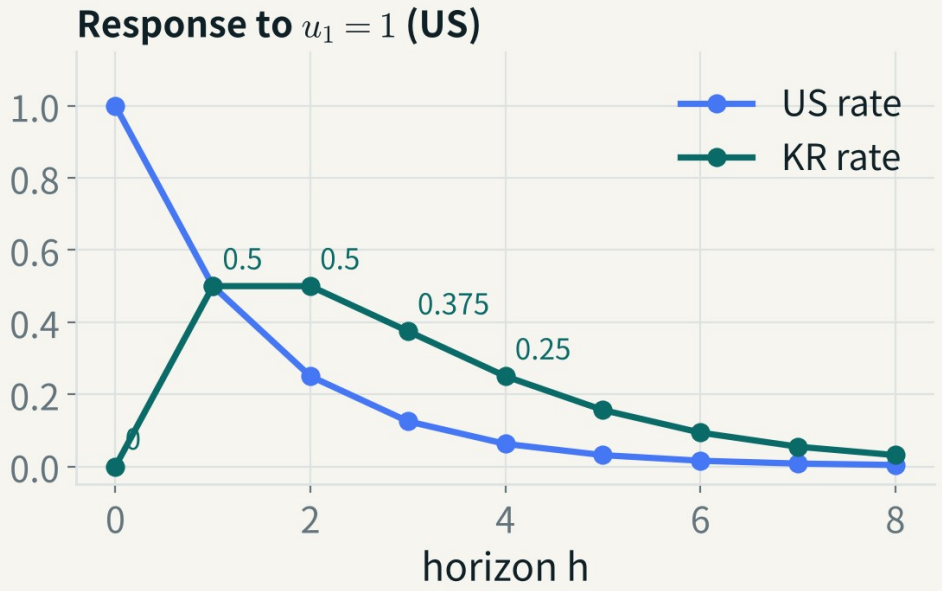
Step 2: a unit US error reaches Korean rates one period later

Column j of Ψ_h = response when only error j is 1 and the other is 0

Definition · response after h periods

$$\Psi_h[i,j] = \frac{\partial y_{i,t+h}}{\partial u_{j,t}}$$

h	US ← u_1	KR ← u_1	US ← u_2	KR ← u_2
0	1	0	0	1
1	0.5	0.5	0	0.5
2	0.25	0.5	0	0.25
3	0.125	0.375	0	0.125
4	0.0625	0.25	0	0.0625



Looking at A alone, Korean rates do not react on the same day (h = 0)

But the errors correlate at 0.6, so u_1 never moves alone in practice

Step 3: the Cholesky split turns correlated errors into independent shocks

Fix the order (US → KR) and the lower-triangular P is unique · ε are independent unit-variance shocks

What we factor

$$\Sigma_u = PP^\top, \qquad P = \begin{bmatrix} p_{11} & 0 \\ p_{21} & p_{22} \end{bmatrix}$$

Result

$$P = \begin{bmatrix} 1 & 0 \\ 0.6 & 0.8 \end{bmatrix}, \qquad u_t = P\,\varepsilon_t, \quad \mathrm{Var}(\varepsilon_t) = I$$

By hand · match PPᵀ to Σ entry by entry

$$\begin{aligned} p_{11}^2 = 1 \quad &\Rightarrow \quad p_{11} = 1 \\ p_{21}\,p_{11} = 0.6 \quad &\Rightarrow \quad p_{21} = 0.6 \\ p_{21}^2 + p_{22}^2 = 1 \quad &\Rightarrow \quad p_{22} = \sqrt{1 - 0.36} = 0.8 \end{aligned}$$

Reading P

- A 1-s.d. US shock moves Korean rates by 0.6 the same day
- A Korean-own shock moves only Korean rates, by 0.8
- 0.36 + 0.64 = 1, so the variance is preserved

The 0.6 correlation is assigned to the same-day effect of the US shock, ordered first

Step 4: orthogonal responses are the matrix powers times P

Entry (i, j) of Θ_h = response of variable i, h periods after a 1-s.d. shock j

Orthogonal IRF · Step 1 rule times Step 3 P

$$\Theta_h = A^h P = 0.5^h \begin{bmatrix} 1 & 0 \\ h & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0.6 & 0.8 \end{bmatrix} = 0.5^h \begin{bmatrix} 1 & 0 \\ h+0.6 & 0.8 \end{bmatrix}$$

h	US ← US shock	KR ← US shock	US ← KR shock	KR ← KR shock
0	1	0.6	0	0.8
1	**0.5**	**0.8**	**0**	**0.4**
2	0.25	0.65	0	0.2
3	0.125	0.45	0	0.1
4	0.0625	0.2875	0	0.05

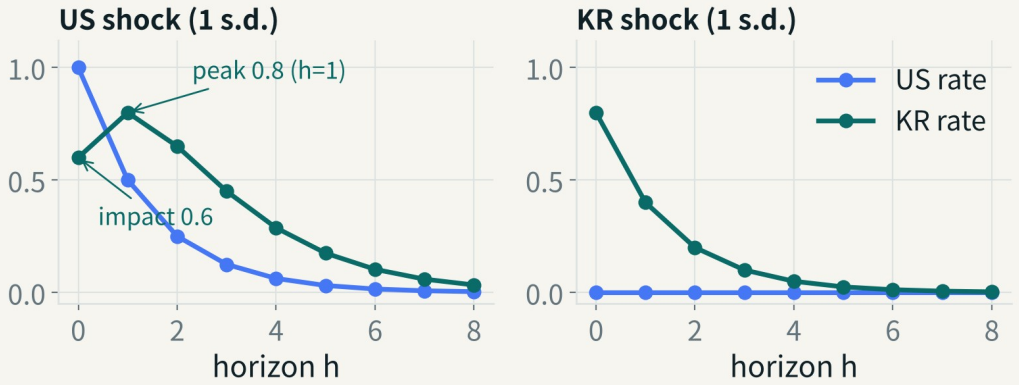
h = 1, KR ← US shock, by hand

$$\theta_{21,1} = \underbrace{0.5 \times 1}_{\text{US lag}} + \underbrace{0.5 \times 0.6}_{\text{own persistence}} = 0.8$$

0.8 after one period = 0.5 passed on from US rates + 0.3 carried by Korean rates themselves

Step 5: the Korean response peaks after one period and then fades

Left: US shock · right: Korean shock (both 1 s.d.) · Step 4 table plotted



Cumulative response · the total effect over all horizons

$$\sum_{h=0}^{\infty} 0.5^h (h + 0.6) = \underbrace{\sum_{h} h\, 0.5^h}_{=2} + 0.6 \underbrace{\sum_{h} 0.5^h}_{=2} = 3.2$$

0.6 on impact, 0.8 one period later, then decay: a hump shape

One US shock moves Korean rates by 3.2 in total

Step 6: reordering the variables changes who owns the same-day effect

Redo the Cholesky split with the same Σ but the order (KR, US)

P with the order reversed

$$P^{\text{rev}} = \begin{bmatrix} 1 & 0 \\ 0.6 & 0.8 \end{bmatrix} \quad \text{(order: KR, US)}$$

Order	US shock → KR (impact)	KR shock → US (impact)
US → KR	0.6	0
KR → US	0	0.6

What changes

- The 0.6 correlation always goes to the shock of the variable ordered first
- A (the lagged effects) is unchanged; only the h = 0 split and the paths after it change
- IRF and FEVD conclusions can depend on the ordering

The order is an assumption about who moves first within the day; justify it economically

02

Forecast Error Variance Decomposition

Steps 7–8 · what share of the forecast error comes from each shock

The H-step forecast error is a weighted sum of future independent shocks

Future shocks are unknown at t, so they form the error; independence splits the variance cleanly

Forecast error · weights are the Step 4 Θ

$$y_{t+H} - \hat{y}_{t+H|t} = \sum_{h=0}^{H-1} \Theta_h \, \varepsilon_{t+H-h}$$

Variance of the Korean forecast error

$$\mathrm{MSE}_2(H) = \sum_{h=0}^{H-1} \left(\theta_{21,h}^2 + \theta_{22,h}^2 \right)$$

FEVD · share of shock j

$$\omega_{2j}(H) = \frac{\sum_{h=0}^{H-1} \theta_{2j,h}^2}{\mathrm{MSE}_2(H)}$$

FEVD = squared responses to each shock, summed over horizons, divided by the total

With the IRF table in hand, FEVD is just squaring and adding

Step 7: square the responses and stack them to get each shock's contribution

Korean rate (variable 2) · responses from the Korean rows of the Step 4 table

h	US-shock resp.²	KR-shock resp.²	Cum. · US	Cum. · KR	Total = MSE
0	$0.6^2 = 0.36$	$0.8^2 = 0.64$	0.36	0.64	1.00
1	**$0.8^2 = 0.64$**	**$0.4^2 = 0.16$**	**1.00**	**0.80**	**1.80**
2	$0.65^2 = 0.4225$	$0.2^2 = 0.04$	1.4225	0.84	2.2625
3	$0.45^2 = 0.2025$	$0.1^2 = 0.01$	1.625	0.85	2.475

Example: for H = 2 add the rows h = 0 and 1

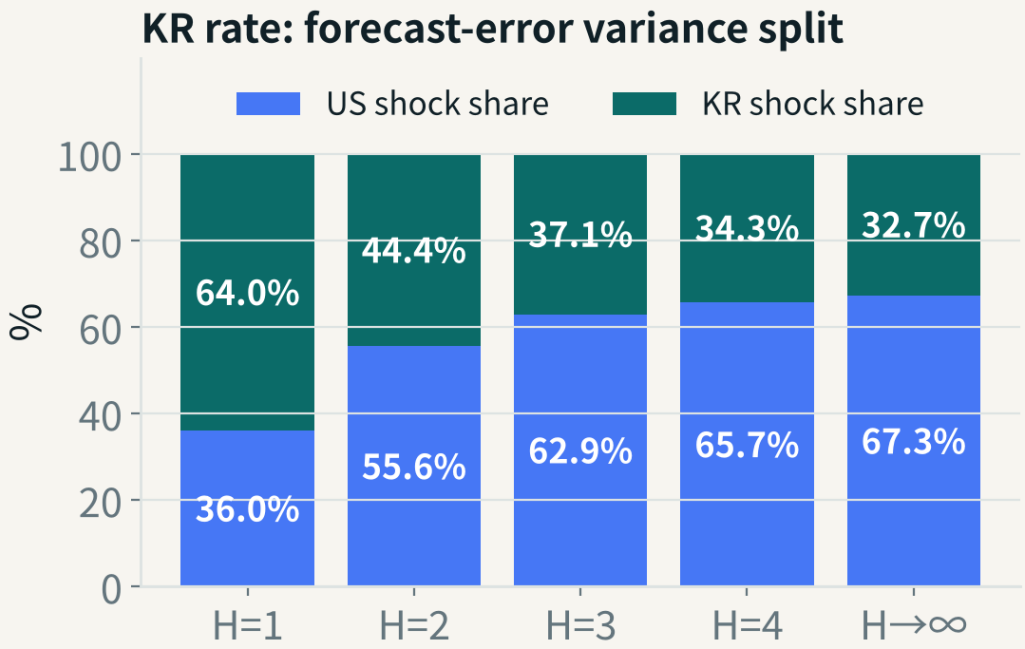
$$\omega_{21}(2) = \frac{0.36 + 0.64}{(0.36 + 0.64) + (0.64 + 0.16)} = \frac{1.00}{1.80} = 55.6\%$$

An H-step forecast uses the H rows h = 0 … H−1

Step 8: the longer the horizon, the larger the US share of Korean forecast errors

Step 7 cumulative sums divided by the MSE · each row sums to 100%

H	US shock share	KR shock share
1	0.36 / 1.00 = 36.0%	64.0%
2	**1.00 / 1.80 = 55.6%**	**44.4%**
3	1.4225 / 2.2625 = 62.9%	37.1%
4	1.625 / 2.475 = 65.7%	34.3%
∞	67.3%	32.7%



Short run: Korean shocks dominate (64%); long run: US shocks do (67%)

US rates are 100% own-shock at every H: both A and the ordering bar Korea from moving the US

statsmodels reproduces the hand calculations exactly

Feed A and Σ to VARProcess to get Ψ and Θ; with real data use the same methods on VAR(df).fit()

```
import numpy as np
from statsmodels.tsa.vector_ar.var_model import VARProcess
A = np.array([[0.5, 0.0], [0.5, 0.5]])
S = np.array([[1.0, 0.6], [0.6, 1.0]])
p = VARProcess(A[None], None, S)
p.ma_rep(4)        # Psi_h   (Step 2)
p.orth_ma_rep(4)   # Theta_h (Step 4)
# real data
res = VAR(df).fit(1)
res.irf(10).plot(orth=True)
res.fevd(10).summary()
```

Check

- orth_ma_rep(4)[1] → 0.5, 0, 0.8, 0.4, matching Step 4
- orth_ma_rep(4)[4] → 0.0625, 0, 0.2875, 0.05
- fevd(2), Korean row → 0.556, 0.444, matching Step 8
- plot(orth=True) draws the Step 5 chart

The code takes the order from the DataFrame columns, so column order is the Step 6 assumption

Exercises: change one assumption at a time and see what moves

All solvable with the numbers in this workbook · answers on the next page

Problem 1

If the correlation is 0

- With Σ = I, what US share in the 2-step FEVD of Korean rates?
- Compare with 55.6% and interpret

Problem 2

If the order flips

- With order (KR, US), what KR share in the 1-step US error?
- How does it differ from Step 3?

Problem 3

Why the peak at h = 1?

- Explain why the Korean response peaks at h = 1,
- not h = 0, using two channels

After each number, write one sentence on why it comes out that way

Answers: correlation and ordering shape both the FEVD and the IRF

Problems 1–2 reuse Steps 6–8; Problem 3 reuses Step 4

Problem 1 · 16.7%

$$\Sigma_u = I: \quad \theta_{21} = (0,\ 0.5), \quad \theta_{22} = (1,\ 0.5) \ \Rightarrow \ \omega_{21}(2) = \frac{0.25}{0.25 + 1.25} = 16.7\%$$

Without the same-day 0.6, the US share drops from 55.6% to 16.7%

Problem 2 · 36%

$$\text{US} = 0.6\,\varepsilon_{KR} + 0.8\,\varepsilon_{US} \ \Rightarrow \ \omega_{US \leftarrow KR}(1) = \frac{0.36}{0.36 + 0.64} = 36\%$$

The same 0.6 now counts as a Korean shock; the ordering changes the conclusion

Problem 3 · 0.5 + 0.3

$$\theta_{21,1} = \underbrace{0.5 \times 1}_{\text{US lag}} + \underbrace{0.5 \times 0.6}_{\text{own persistence}} = 0.8$$

0.5 passed on from US rates plus 0.3 from Korea's own persistence of the 0.6

Remember: IRF and FEVD reflect Σ and the ordering assumption, not A alone